Supporting Information

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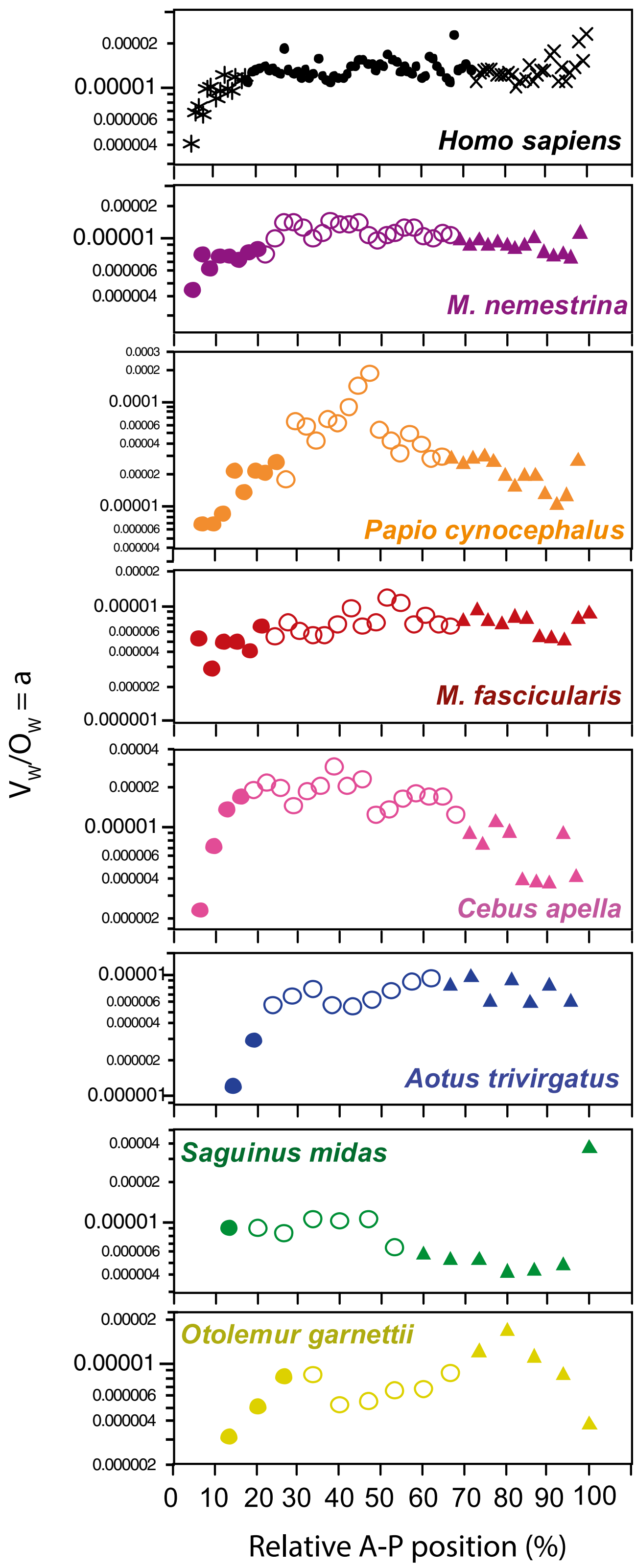

Fig. S1. Smaller average fiber caliber in cortex anterior to corpus callosum, defined here as prefrontal cortex. The ratio between V_{WM} and O_{WM}, which approximates the average fiber caliber, is shown for each coronal section along the A-P axis. Filled circles, prefrontal cortex (coronal sections anterior to corpus callosum); triangles, occipital cortex (posterior third of V_{GM} along the A-P axis); and unfilled circles, intermediate regions. Species are shown in order of decreasing gray matter volume.

Table S1. Percent volume and number of cortical neurons in prefrontal region

Species	% V_{GM}	% V_{WM}	% O_{WM}	% neurons
O. garnettii	9.2	3.7	7.6	8.3
S. midas	7.4	3.6	2.8	6.4
A. trivirgatus	9.2	1.9	7.4	7.8
C. apella	6.6	3.0	3.6	4.2
M. fascicularis	7.7	4.4	6.5	6.9
P. cynocephalus	13.5	8.0	14.4	16.2
M. nemestrina	7.5	4.6	6.8	7.8
Homo sapiens	10.0	5.5	7.2	7.8

% V_{GM}, percentage of gray matter volume contained in prefrontal region; % V_{WM}, percentage of white matter volume contained in prefrontal region; % O_{WM}, percentage of all other (nonneuronal) cells in the white matter contained in prefrontal region; and % neurons, percentage of all cortical neurons contained in prefrontal region.

Table S2. Slopes of allometric relationships for raw data and phylogenetically independent contrasts

			Raw data			Phylogenetically independent contrasts		
Dependent variable	Independent variable	Region	Slope ± SE	r^2	*P* value	Slope ± SE	r^2	*P* value
Neuron density	V_{GM}	PF	−0.256 ± 0.068	0.738	0.0132	−0.328 ± 0.073	0.801	0.0065
		Int	−0.315 ± 0.088	0.720	0.0157	−0.324 ± 0.113	0.620	0.0355
		Occ	−0.313 ± 0.112	0.610	0.0381	−0.289 ± 0.147	0.435	0.1070
V_{GM}	Neurons	PF	1.290 ± 0.118	0.960	0.0001	1.405 ± 0.153	0.944	0.0002
		Int	1.349 ± 0.173	0.924	0.0006	1.297 ± 0.218	0.876	0.0019
		Occ	1.285 ± 0.210	0.872	0.0017	1.158 ± 0.240	0.824	0.0047
V_{WM}	Other cells, WM	PF	1.176 ± 0.222	0.848	0.0032	1.206 ± 0.303	0.760	0.0105
		Int	1.241 ± 0.254	0.827	0.0045	1.310 ± 0.347	0.740	0.0130
		Occ	1.137 ± 0.211	0.853	0.0030	1.158 ± 0.198	0.837	0.0021
V_{WM}	Neurons	PF	1.656 ± 0.298	0.860	0.0026	1.765 ± 0.415	0.783	0.0081
		Int	1.644 ± 0.208	0.926	0.0005	1.695 ± 0.258	0.896	0.0012
		Occ	1.557 ± 0.260	0.878	0.0019	1.553 ± 0.271	0.868	0.0023
Neurons, PF	Neurons, NPF		0.917 ± 0.220	0.776	0.0088	0.779 ± 0.210	0.733	0.0139
Other cells, PF WM	Other cells, NPF WM		0.966 ± 0.256	0.739	0.0131	0.914 ± 0.237	0.749	0.0119

All functions calculated without human data points ($n = 7$). Neuron density in neurons per cubic millimeter. V_{GM}, volume of cortical gray matter (cubic millimeters). V_{WM}, volume of subcortical white matter (cubic millimeters). Int, intermediate region; NPF, nonprefrontal region; Occ, occipital region; and PF, prefrontal region.